FootSense PRD

1. Idea

FootSense is a smart insole system designed to understand how a person walks and translate that into clear, meaningful feedback. Rather than simply tracking steps, it focuses on the quality of movement by analyzing pressure distribution, gait patterns, and balance.

The system consists of pressure sensors embedded within a gel insole, connected to a compact external module that handles processing and communication. Data is transmitted to a mobile device, where it is analyzed to provide insights that help improve posture, prevent injury, and refine movement.

2. Goal of the Project

The objective of FootSense is to address everyday issues in walking and posture that often go unnoticed. Many individuals develop inefficient or uneven movement patterns over time, which can lead to discomfort, injury, or reduced performance.

This project aims to create a system that brings awareness to these issues by continuously monitoring how weight is distributed across the foot and how each step is taken. It is also intended to support injury prevention and recovery by identifying irregular gait patterns early and guiding users toward safer movement habits.

In addition, FootSense seeks to enhance training and performance by providing detailed insights into movement quality. By combining sensor data with intelligent analysis, the system delivers practical suggestions that users can apply immediately. The design prioritises usability, ensuring that the insole remains comfortable while housing the main electronics externally.

3. System Architecture Overview

- **Insole (inside shoe):** Pressure sensors embedded in gel
- **External module:** Processing, motion sensing, and communication in a compact enclosure
- **Mobile device:** Data visualization and intelligent analysis

4. Sensors and Components

Pressure Sensing (Insole):

- Multiple Force Sensitive Resistors (FSRs) distributed across the foot
  Example: Interlink FSR 402

Motion and Gait Detection (External Module):

- IMU (accelerometer and gyroscope)
  Example: MPU6050

Processing and Communication:

- ESP32

Actuation and Feedback (Optional):

- L298N Motor Driver Module
- Vibration motor for alerts

Other Components:

- Rechargeable Li-Po battery
- Voltage regulation circuitry
- Connecting wires between insole and module

5. 3D Printed Parts

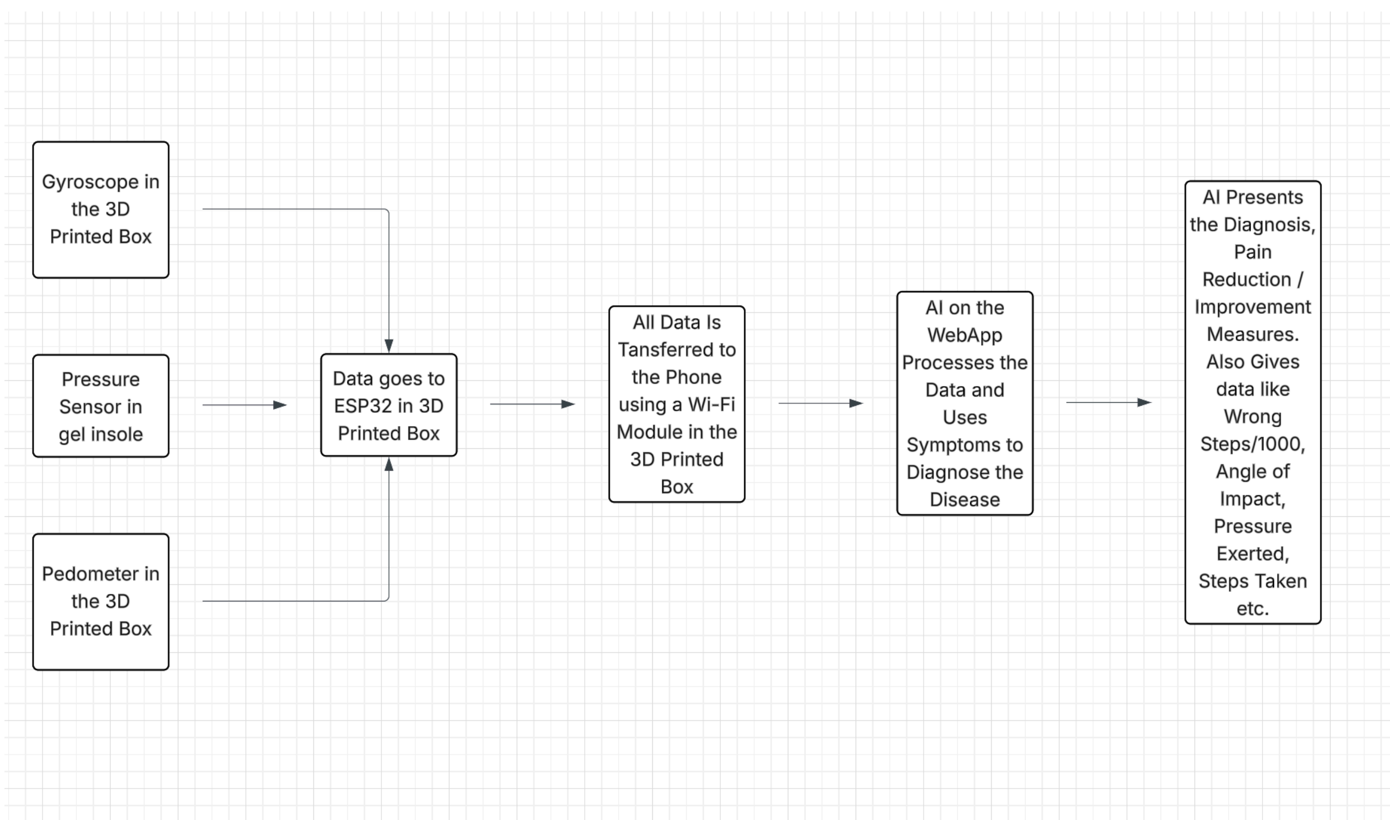
External Electronics Enclosure
A rigid and compact casing attached to the shoe or ankle, designed to securely house the ESP32, IMU, battery, and PCB. It should be durable, lightweight, and easy to access for maintenance.

Insole Sensor Support Layer
A thin internal structure that holds the pressure sensors in fixed positions within the gel insole, ensuring consistent readings during movement.

Wire Routing Channel
A protected pathway that guides wires from the insole to the external module, minimizing stress and preventing damage during regular use.

Protective Sensor Layer
A flexible top layer that shields the sensors from moisture and repeated pressure while maintaining sensitivity and comfort.

6. System Workflow (Flowchart Text)



[Idea & Planning]

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[Research Components]

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[Design Circuit]

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[Assemble Sensors]

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[Write & Test Code]

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[Test Sensor Readings]

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[Analyse Data (Steps, Pressure, Balance)]

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[Add Output (Bluetooth)]

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[Design Insole Structure (CAD)]

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[3D Print & Assemble Hardware]

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[Final Testing & Improvements]

7. AI Output and User Insights

The system presents data in a clear and actionable manner. It calculates the number of incorrect steps per 1000 steps and identifies specific issues such as uneven pressure distribution, excessive impact, or imbalance. It also highlights patterns over time and correlates them with user-reported symptoms such as discomfort or fatigue. Based on this analysis, FootSense provides practical recommendations to improve walking technique, enhance balance, and reduce strain.

8. Success Criteria

The project is considered successful if it can reliably capture pressure and movement data, accurately identify irregular walking patterns, and present insights that are easy to understand and apply. The system should remain comfortable to wear, maintain stable wireless communication, and function consistently as a complete, demonstrable prototype.